GOVERNMENT OF KARNATAKA
DEPARTMENT OF SCHOOL EDUCATION (PRE-UNIVERSITY)
18TH CROSS, MALLESHWARAM, BENGALURU – 560 012

CHAPTERWISE MULTIPLE CHOICE QUESTIONS FOR COMPETATIVE EXAM

SUBJECT: II PUC CHEMISTRY

NAME OF THE CHAPTER: UNIT: 04-d AND f BLOCK ELEMENTS

1. **The d-block elements:**
   - The d block elements are called transitional elements because their properties are in between s block and p block elements properties.
   - Zinc (Zn), Cadmium (Cd) and Mercury (Hg) of group 12 have full d^{10} configuration in their ground state as well as in their common oxidation states and hence, are not regarded as transition metals.
2. **Electronic Configurations of the d-Block Elements**
   - In general the electronic configuration of these elements is [Noble gas] (n-1) $d^{1–10}$ $ns^{1–2}$.
3. **Meting point and boiling point:**
   - The high melting points of these metals are due to the involvement of greater number of electrons from (n-1)d in addition to the ns electrons in the inter atomic metallic bonding.
   - The melting point of transitional metals increases upto d^5 configuration till the middle of the series then decreases because the number of unpaired electrons increases upto d^5 configuration and then decreases because of pairing of electrons. So, the strength of metallic bond increases and then decreases.
4. **Enthalpy of atomization:**
   - Transitional metals exhibit higher enthalpy of atomization because of presence of higher number of unpaired electrons in strong inter atomic metallic bond.
   - The enthalpy of atomization of Zn in 3d series is lowest because of absence of un paired electrons, metallic bond is weak.
5. **Variation in Atomic and Ionic Sizes of Transition Metals**
   - Atomic size of transitional metals decreases with increase in atomic number due to increase in effective nuclear charge.
   - The decrease in atomic size becomes small after middle of the series because after middle of the series, effective nuclear charge is compensated by shielding effect of d electrons.
   - The atomic size of second series (4d series) and Third series (5d series) are almost same because of Lanthanoid contraction.
   - The atomic size of Zr and Hf are identical due to Lanthanoid contraction.
6. **Ionisation Enthalpies:**
   - Ionisation enthalpies of transitional metals increases along the period due to increase in effective nuclear charge. Here variation is not regular.

- Second ionization enthalpies of Cr and Cu are sufficiently higher than their neighbours because of their stable configurations (d^5 and d^{10})
- First ionization energy of Cr is lower because removal of one electron produces extra stable d^5 configuration.

7. Oxidation States:

- Transitional metals shows variable oxidation state because of comparable energies of (n-1) d and ns orbitals. So they lose electrons from both (n-1) d and ns orbitals.
- 3d series elements shows variable oxidation states because of comparable energies of 3d and 4s orbitals.

8. Magnetic Properties: Paramagnetism is expressed in magnetic moment (μ) (Spin only magnetic moment) and measured in Bohr Magneton (BM) units.

$\mu = \sqrt{n(n+2)}$ Here n is the number of unpaired electrons.

9. Formation of Coloured Ions: Transitional metals forms coloured compounds because of (i) Presence of unpaired electrons. (ii) d-d transition of electrons in the visible region.

10. Formation of Complex Compounds:

Transitional metals form complex compounds because of

- Presence of d orbitals for bond formation.
- High charge and small size of metal atom (High polarizing power)

11. Catalytic Properties:

Transitional metals are good catalysts because of

- Their ability to adopt variable oxidation states.
- Their ability to form complexes
- They provide large surface area for catalytic activity.

12. Formation of Interstitial Compounds:

Interstitial compounds: Interstitial compounds are those which are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals. Examples: TiC, Mn_4N, Fe_3H, $VH_{0.56}$ and $TiH_{1.7}$, etc.

- They have high melting points, higher than those of pure metals.
- They are very hard, some borides approach diamond in hardness.
- They retain metallic conductivity.
- They are chemically inert.

13. Alloys: Alloys are homogeneous mixture of two or more than two metals or metals and nonmetals. Transitional metals can form alloys because they have similar size and in the crystal lattice one metal can be replaced by other.

14. Oxides and Oxoanions of Metals:

- All transitional metals except Sc forms oxides which are generally basic in nature.
- Higher oxidation state oxides like Mn_2O_7 are covalent and acidic in nature.
- Some metal oxides are amphoteric in nature (Cr_2O_3 , V_2O_5)

15. Manufacturing of Potassium dichromate:

$$4\ FeCr_2O_4 + 8\ Na_2CO_3 + 7\ O_2 \rightarrow 8\ Na_2CrO_4 + 2\ Fe_2O_3 + 8\ CO_2$$

$$2Na_2CrO_4 + 2H^+ \rightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$$

$$Na_2Cr_2O_7 + 2KCl \rightarrow K_2Cr_2O_7 + 2NaCl$$

$$2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$$

$$Cr_2O_7^{2-} + 2OH^- \rightarrow 2CrO_4^{2-} + H_2O$$

Sodium and potassium dichromates are strong oxidising agents. In acidic solution, its oxidising action can be represented as follows:

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$$

Thus, acidified potassium dichromate will oxidise iodides to iodine, sulphides to sulphur, tin(II) to tin(IV) and iron(II) salts to iron(III).

$6I^- \rightarrow 3I_2 + 6e^-$; $3Sn^{2+} \rightarrow 3Sn^{4+} + 6e^-$

$3H_2S \rightarrow 6H^+ + 3S + 6e^-$; $6Fe^{2+} \rightarrow 6Fe^{3+} + 6e^-$

- $Cr_2O_7^{2-} + 14H^+ + 6I^- \rightarrow 2Cr^{3+} + 7H_2O + 3I_2$
- $Cr_2O_7^{2-} + 14H^+ + 3Sn^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 3Sn^{4+}$
- $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 6Fe^{3+}$
- $Cr_2O_7^{2-} + 8H^+ + 3H_2S \rightarrow 2Cr^{3+} + 7H_2O + 3S$

16. Manufacturing of Potassium permanganate:

$$2MnO_2 + 4KOH + O_2 \rightarrow 2K_2MnO_4 + 2H_2O$$

$$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$$

Action of heat : When heated Potassium permanganate decomposes at 513 K.

$$2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$$

17. Oxidising reactions of $KMnO_4$:

- **In acid solutions:**
  - $10I^- + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5I_2$
  - $5Fe^{2+} + MnO4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$
  - $5C_2O_4^{2-} + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 10CO_2$
  - $H_2S \longrightarrow 2H^+ + S^{2-}$
  - $5S^{2-} + 2MnO_4^{--} + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5S$
  - $5SO_3^{2-} + 2MnO_4^- + 6H^+ \rightarrow 2Mn^{2+} + 3H_2O + 5SO_4^{2-}$
  - $5NO_2^- + 2MnO_4^- + 6H^+ \rightarrow 2Mn^{2+} + 5NO_3^- + 3H_2O$
- **In neutral or faintly alkaline solutions:**
  - $2MnO_4^- + H_2O + I^- \rightarrow 2MnO_2 + 2OH^- + IO_3^-$
  - $8MnO_4^- + 3S_2O_3^{2-} + H_2O \rightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$
  - $2MnO_4^- + 3Mn^{2+} + 2H_2O \rightarrow 5MnO_2 + 4H^+$

THE INNER TRANSITION ELEMENTS (f – BLOCK)

The f block elements: The elements in which last electrons are filling to (n-2) [Pre-penultimate] f sub shell are called f – block elements.

- There are two series in f block elements. They are (1) Lanthanoids (2) Actinoids

They are called as inner transitional elements because last electrons are filling to innermost(n-2) f subshell.

They kept separately in periodic table because they have similarity in properties along the row and not along the columns (groups).

The general electronic configuration of f – block elements is [Noble gas](n-2)f^{1-14}(n-1)$d^{0-1}ns^2$.

I. Lanthanoids: The elements in which added electrons fills into 4f [$4f^{1-14}$] subshell are called Lanthanoids.

- The elements from Cerium(Ce)[Z=58] to Lutetium(Lu)[Z=71] belongs to Lanthanoids.

1. Electronic Configuration:

The general electronic configuration of lanthanoids is [Xe]$4f^{1-14}5d^{0-1}6s^2$

2. Lanthanoid contraction: The gradual decrease in the atomic and ionic radius of lanthanoids with increase in their atomic number is called lanthanoid contraction.

- Lanthanoid contraction is due to poor shielding effect of 4f electrons.

3. Consequences of Lanthanoid contraction:

- Radius of 4d series (second series) and 5d series (Third series) elements are identical.
- Radii of Zirconium (Zr)[160 pm] and Hafnium (Hf) [159 pm] are identical.

4. Oxidation States:

Lanthanoids shows variable oxidation state due to comparable energies of 4f, 5d and 6s orbitals.

Common oxidation state of lanthanoids is +3. Cerium shows +4 oxidation state because it can attain stable noble gas (Xe) configuration.

5. General Characteristics:

- The lanthanoid ions other than the f^{0} type (La^{3+} and Ce^{4+}) and the f^{14} type (Yb^{2+} and Lu^{3+}) are all paramagnetic. The paramagnetism rises to maximum in Neodymium.
- The best single use of the lanthanoids is for the production of alloy steels for plates and pipes. A well known alloy is mischmetal which consists of a lanthanoid metal (~ 95%) and iron (~ 5%) and traces of S, C, Ca and Al. A good deal of mischmetal is used in Mg-based alloy to produce bullets, shell and lighter flint. Mixed oxides of lanthanoids are employed as catalysts in petroleum cracking. Some individual Ln oxides are used as phosphors in television screens and similar fluorescing surfaces.

6. Chemical reactions of the lanthanoids:

- Lanthanoids(Ln) burns with Oxygen to form oxides of the form Ln_2O_3

$$4\ Ln + 3\ O_2 \rightarrow 2\ Ln_2O_3.$$

- Lanthanoids(Ln) heated with Carbon to form carbides of the form Ln_3C, Ln_2C_3, LnC_2

$$3\ Ln + C \rightarrow Ln_3C$$

$2\ Ln + 3\ C \rightarrow Ln_2C_3$

$Ln + 2\ C \rightarrow LnC_2$

- Lanthanoids(Ln) reacts with halogen(X) to form halides.

$Ln + 3\ X \rightarrow LnX_3$

- Lanthanoids(Ln) reacts with Nitrogen to form nitrides.

$2Ln + N_2 \rightarrow 2LnN$

- Lanthanoids(Ln) reacts with Sulphur to form Sulphides .

$2Ln + 3S \rightarrow Ln_2S_3$

- Lanthanoids(Ln) reacts with water to form hydroxides to liberating hydrogen gas.

$2Ln + 6\ H_2O \rightarrow 2\ Ln(OH)_3 + 3\ H_2$

- Lanthanoids(Ln) reacts with dilute acids and liberates hydrogen gas.

$Ln +$ Dilute acids $\rightarrow$ Liberates hydrogen.

II. The Actinoids: The elements in which added electrons fills into 5f [$5f^{1-14}$] subshell are called Actinoids. The elements from Thorium (Th)[Z=90] to Lawrencium (Lr)[Z=103] belongs to Actinoids.

1. **Electronic Configuration:** The general electronic configuration of Actinoids is $[Rn]5f^{1-14}6d^{0-1}7s^2$
2. **Actinoid contraction:** The gradual decrease in the atomic and ionic radius of actinoids with increase in their atomic number is called actinoid contraction.
   - Actinoid contraction is due to poor shielding effect of 5f electrons.
   - Actinoid contraction is greater than lanthanoid contraction because shielding effect of 5f electrons are comparatively less than that of shielding effect of 4f electrons.
3. **Oxidation States:**

   Actinoids shows variable oxidation state due to comparable energies of 5f , 6d and 7s orbitals.

   Common oxidation state of actinoids is +3.
4. **General Characteristics :**(1)

- The actinoid metals are all silvery in appearance.
- Study of actinoids is difficult because
  - All of them radioactive.
  - Many of them have very short half-life.
  - They could be prepared in nanograms.
- Actinoids are more reactive than lanthanoids because 5f electrons of actinoids are less firmly held by nucleus than 4f electrons of lanthanoids.

5. **Some Applications of d- and f-Block Elements:**
   - Iron and steel are used as construction material.
   - Some oxides like TiO used in pigment industry.
   - MnO_2 used in dry cell, Zn and Ni/Cd also required in battery cells.
   - Cu/Ni alloys used in making coins.
   - Ziegler-Natta catalyst [$TiCl_4$ with $(CH_3)_3Al$] used in manufacturing of Polythene.

- AgBr used in photography.
- Transitional metals are used as catalysts in many industrial productions.

The d-block elements

1. Which of the following is non typical transitional element?

(a) Cr (b) Mn

(c) Fe (d)Zn

Electronic Configurations of the d-Block Elements

2. The electronic configuration of Cu(II) is $3d^9$ whereas that of Cu(I) is $3d^{10}$. Which of the following is correct?

a) Cu (II) is more stable

b) Cu (II) is less stable

c) Cu(I) and Cu (II) are equally stable

d) Stability of Cu(I) and Cu (II) depends on nature of copper salts

3. The electron configuration of element with Z = 24 is

a) $3d^4\ 4s^2$ b) $3d^5\ 4s^1$

c) $3d^6\ 4s^1$ d) none of the above

4. Which one of the following ions has electronic configuration $[Ar]3d^6$?

(a) Ni^{3+} (b) Mn^{3+}

(c) Co^{3+} (d) Fe^{3+}

Meting point and boiling point

5. In general, the melting and boiling points of transition metals

(a) increases gradually across the period from left to right.

(b) decreases gradually across the period from left to right.

(c) first increases till the middle of the period and then decreases towards the end.

(d) first decreases regularly till the middle of the period and then increases towards the end.

6. Among the transition elements the element with lower melting point belongs to

a) Group 3 b) Group 11

c) Group 6 d) Group 12

Enthalpy of atomization

7. Transition elements exhibit higher enthalpies of atomization

a) Because they exhibit variable oxidation state

b) Because of availability of higher number unpaired electron for metallic bonding

c) Because of magnetic nature

d) Because they are radioactive

8. The $E_{M2+/M}$ value for copper is positive (+ 0.34V). What is possible reason for this?

a) Copper has a high value of atomization enthalpy and low hydration enthalpy.

b) Copper has small size

c) Because of Lanthanoid contraction

d) Because of variable oxidation state

9. In the series Sc (Z = 21) to Zn (Z = 30), the enthalpy of atomization of zinc is the lowest, i.e., 126 kJ mol–1, because

a) Zinc shows only one oxidation state
b) Zinc has small size
c) Zinc has one unpaired electron
d) Zinc has no unpaired electron

Ionisation Enthalpies

10. The second ionisation enthalpy value of Cr and Cu is exceptionally high because,

a) Both contains 1 unpaired electron
b) No unpaired electrons
c) M^+ ions have the d^5 and d^{10} configurations respectively.
d) Because of Lanthanoid contraction

Oxidation States

11. Of the following outer electronic configuration, highest oxidation state is achieved by

(a) (n-1) d^3ns^2
(b) (n-1) d^5ns^1
(c) (n-1) d^8ns^2
(d) (n-1) d^5ns^2

12. The element that usually does not show variable oxidation states is

(a) Sc
(b) Cu
(c) Ti
(d) V

13. The pair of elements which show + 1 oxidation state

(a) Cr & Zn
(b) Fe & Zn
(c) Cr & Cu
(d) Cu & Zn

14. The element showing oxidation state +7 is

a) Cr
b) Mn
c) Co
d) V

15. Of the following elements which is not expected to display an oxidation state of +6 in any of its compounds?

a) Ti
b) Cr
c) Mn
d) W

16. A transition metal exists in its highest oxidation state. It is expected to behave as

(a) an oxidising agent
(b) a chelating agent
(c) a reducing agent
(d) a central metal in a co-ordination

17. Which one of the following is correct for all elements from Sc to Cu?

a) The lowest oxidation state shown by them is +2.
b) 4s orbital is completely filled in the ground state.
c) *3d* orbital is not completely filled in the ground state.
d) The ions in +2 oxidation states are paramagnetic.

18. Match the followings aspects with the respective metal.

Aspects
Metal

A. The metal which reveals a maximum number of oxidation states	I. Scandium
B. The metal although placed in 3d block is considered not as a transition element	II. Copper
C. The metal which does not exhibit variable oxidation states	III. Manganese
D. The metal which in +1 oxidation state in aqueous solution undergoes disproportionation	IV. Zinc

Select the correct option.

	A	B	C	D
a)	I	IV	II	III
b)	III	IV	I	II
c)	III	I	IV	II
d)	III	V	I	III

19. Copper is a d-block element because

a) It contains partially filled d-orbital in its ground state

b) It contains partially filled d-orbital in its stable oxidation state

c) It contains partially filled d-orbital both in its ground state & stable oxidation state

d)It doesn't contain partially filled d-orbital in its ground state

20. Which of the following has the maximum number of unpaired electrons?

a) Ti^{2+} b) Fe^{2+}

c) Cr^{+} d) Cu^{+}

Magnetic Properties

21. Match the metal ions given in Column I with the spin magnetic moments of the ions given in Column II and assign the correct code

Column I	Column II
1. Co^{3+}	i. $\sqrt{8}$
2. Cr^{3+}	ii. $\sqrt{35}$
3. Fe^{3+}	iii. $\sqrt{3}$
4. Ni^{2+}	iv. $\sqrt{24}$
	v. $\sqrt{15}$

Select the correct option.

	1	2	3	4
a)	iv	i	ii	iii
b)	i	ii	iii	iv
c)	iv	v	ii	i
d)	iii	v	I	ii

22. The calculated spin only magnetic moment of Cr^{2+} ion is

(a) 4.90 BM (b) 2.84 BM

(b) 5.92 BM (d) 3.87 BM

23. Which one of the following will exhibit minimum paramagnetic behavior?

(a) $[Fe(H_2O)_6]^{2+}$ (b) $[Co(H_2O)_6]^{2+}$

(c) $[Cr(H_2O)_6]^{2+}$ (d) $[Mn(H_2O)_6]^{2+}$

24. Which one of the following ions exhibits d-d transition and paramagnetism as well?

(a) MnO_4^- (b) $Cr_2O_7^{2-}$

(c) CrO_4^{2-} (d) MnO_4^{2-}

25. Among the following lowest degree of paramagnetism per mole of compound at 298K will be shown by

(a) $MnSO_4.4\ H_2O$ (b) $NiSO_4.6\ H_2O$

(c) $FeSO_4.6\ H_2O$ (d) $CuSO_4.5\ H_2O$

26. In transition series, with increase in atomic number, the paramagnetism

(a) Increases gradually

(b) Decreases gradually

(c) First increases to maximum then decreases

(d) First decreases to minimum and then increases

27. Chloro compound of vanadium has only spin magnetic moment of 1.73 BM. This vanadium chloride has the formula:

a) VCl_2 b) VCl_4

c) VCl_3 d) VCl_5

28. Which of the following ions has maximum magnetic moment

a) Mn^{+2} b) Fe^{+2}

c) Ti^{+2} d) Cr^{+2}

29. Identify the set of paramagnetic ions among the following

a) Ti^{3+},Cu^{2+}, Mn^{3+} b) Sc^{3+}, Ti^{3+}, V^{3+}

c) V^{2+}, Co^{2+}, Zn^{2+} d) Ni^{2+}, Cu^{2+}, Zn^{2+}

30. A magnetic moment of 1.73 BM will be shown by one of the following

(a) $[Cu(NH_3)_4]^{+2}$ (b) $[TiCl_4]^{-2}$

(c) $[Ni(CN)_4]^{-2}$ (d) $[CoCl_6]^{-4}$

31. The value of spin only magnetic moment for one of the following is 2.84 BM. The correct one is

(a) d^5 in strong field (b) d^4 in strong field

(c) d^4 in weak field (d) d^3 in weak field

Formation of Coloured Ions

32. The d-block elements form-coloured ions because

a) They absorb some energy for d- s transition

b) They absorb some energy for p-d transition

c) They absorb some energy for d-d transition

d) They do not absorb any energy

33. Which of the following ions are coloured?

(a) Cu^{+} (b) Zn^{2+}

(c) Ti^{4+} (d) V^{3+}

34. Cuprous ion is colourless while cupric ion is coloured because

(a) cuprous ion has incomplete d-orbital and cupric ion has a complete d-orbital

(b) both have half-filled p and d-orbitals

(c) cuprous ion has a complete d-orbital and cupric ion has an incomplete d- orbital

(d) both have unpaired electrons in the d-orbitals

Formation of Complex Compounds

35. Which of the following is not a condition for complex formation?

(a) Small atomic size (b) High nuclear charge

(c) Variable oxidation states (d) Availability of vacant d orbital

36. Which of the following complex ion is not expected to absorb visible light?

(a) $[Ni(CN)_4]^{-2}$ (b) $[Cr(NH_3)_6]^{+3}$

(c) $[Fe(H_2O)_6]^{+2}$ (d) $[Ni(H_2O)_6]^{+2}$

Formation of Interstitial Compounds

37. Which of the following statements about the interstitial compounds is incorrect?

(a) They retain metallic conductivity

(b) They are chemically reactive

(c) They are much harder than the pure metal

(d) They have higher melting points than the pure metals

38. Which transition metal has lowest density?

(a) Sc (b) Ti

(c) Zn (d) La

39. Which of the following ions has the maximum magnetic moment?

a) Mn^{+2} b) Fe^{+2}

c) Ti^{+2} d) Cr^{+2}

40. Which one of the following does not correctly represent the correct order of the property indicated against it?

a) Ti < V < Cr < Mn: increasing number of oxidation states

b) $Ti^{3+} < V^{3+} < Cr^{3+} < Mn^{3+}$: increasing magnetic moment

c) Ti < V < Cr < Mn: increasing melting points

d) Ti < V < Mn < Cr: increasing 2nd ionization enthalpy

Catalytic Properties

41. The catalytic activity of transition metals and their compounds is mainly due to

a) their magnetic behaviour b) their unfilled *d*-orbitals

c) their ability to adopt variable oxidation state d) their chemical reactivity

42. Identify the incorrect statement.

(a) The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes.

(b) Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.

(c) The oxidation states of chromium in CrO_4^{2-} and $Cr_2O_7^{2-}$ are not the same.

(d) Cr^{2+} (d^4) is a stronger reducing agent than Fe^{2+} (d^6) in water.

Alloys

43. Transition elements form alloys readily because they have

(a)similar properties (b) same outer electronic configurations

(c) similar crystal structure (d) same incompletely filled d-orbitals

44. Bronze is an alloy of

a) Manganese – zinc b) Copper – tin

c) Copper – nickel d) Scandium – Titanium

45. Cu^+ ion is not stable in aqueous solutions, due to

a) It has high hydration enthalpy b) It has low hydration enthalpy

c) small size d) None of the above

Oxides and Oxoanions of Metals

46. The acidic, basic or amphoteric nature of Mn_2O_7, V_2O_5 and CrO are respectively

(a) Acidic, acidic and basic (b) Basic, amphoteric and acidic

(c) Acidic, amphoteric and basic (d) Acidic, basic and amphoteric

47. Which of the following is an amphoteric oxide? CET 2018

(a) V_2O_5, Cr_2O_3 (b) Mn_2O_7, Cr_2O_3

(c) CrO, V_2O_5 (d) V_2O_5, V_2O_4

48. The basic character of the transition metal monoxide follows the order

(a) VO > CrO > TiO > FeO (b) CrO > VO > FeO > TiO

(c) TiO > FeO > VO > CrO (d) TiO > VO > CrO > FeO

Potassium dichromate

49. Which of the following statements is correct when SO_2 is passed through acidified $K_2Cr_2O_7$ solution?

(a) SO_2 is reduced (b) Green $Cr_2(SO_4)_3$ is formed

(c) The solution turns blue (d) The solution is decolorized

50. A mixture of NaCl and $K_2Cr_2O_7$ is heated with Con. H_2SO_4, deep red vapors are formed. Which of the following statements is false? CET 2019

(a) The vapors give a yellow solution with NaOH

(b) The vapors contain CrO_2Cl_2 only

(c) The vapors contain CrO_2Cl_2 and Cl_2

(d) The vapors when passed into lead acetate in acetic acid gives a yellow precipitate

51. How many moles of acidified $K_2Cr_2O_7$ is required to liberate 6 moles of I_2 from an aqueous solution of I^-? CET 2020

(a) 1 (b) 0.25

(c) 0.5 (d) 2

52. In the dichromate anion

(a) 4 Cr – O bonds are equivalent (b) 6 Cr – O bonds are equivalent

(c) All Cr – O bonds are equivalent (d) All Cr – O bonds are not equivalent

Potassium permanganate

53. $KMnO_4$ acts as an oxidising agent in alkaline medium. When alkaline $KMnO_4$ is treated with KI, iodide ion is oxidised to CET 2018

(a) I_2 (b) IO^-

(c) IO_3^- (d) IO_4^-

54. $aMnO_4^- + bS_2O_3^{-2} + H_2O \rightarrow xMnO_2 + ySO_4^{-2} + zOH^-$. a and y respectively are CET 2023

(a) 3; 6 (b) 8;8

(c) 8;3 (d) 8;6

55. In the reaction between moist SO_2 and acidified permanganate solution: CET 2024

(a) SO_2 is oxidised to SO_4^{-2}, MnO_4^- is oxidised to MnO_4

(b) SO_2 is reduced to S, MnO_4^- is reduced to Mn^{+2}

(c) SO_2 is oxidised to SO_3^{-2}, MnO_4^- is oxidised to MnO_4

(d) SO_2 is reduced to H_2S, MnO_4^- is reduced to MnO_2

56. In the reaction between hydrogen sulphide and acidified permanganate solution, CET 2025

(a) H_2S is reduced to S, MnO_4^- is oxidised to Mn^{+2}

(b) H_2S is oxidised to SO_2, MnO_4^- is reduced to MnO_2

(c) H_2S is reduced to SO_2, MnO_4^- is oxidised to Mn^{+2}

(d) H_2S is oxidised to S, MnO_4^- is reduced to Mn^{+2}

57. $aC_2O_4^{-2} + bMnO_4^- + c\ H^+ \rightarrow xMn^{2+} + yH_2O + zCO_2$. a and x respectively are CET 2026

(a) 5,2 (b) 4,1

(c) 3,2 (d) 4,2

58. $KMnO_4$ is a strong oxidizing agent in acid medium. To provide acid medium H_2SO_4 is used instead of HCl this is because

(a) H_2SO_4 is stronger acid than HCl (b) HCl is oxidized by $KMnO_4$ to Cl_2

(c) H_2SO_4 is a dibasic acid (d) rate is faster in the presence of H_2SO_4

59. Aqueous solution of a salt A forms a dense white precipitate with $BaCl_2$ solution. The precipitate dissolves in dilute HCl to produce a gas B which decolorizes acidified $KMnO_4$ solution. A and B respectively are CET 2020

(a) $BaSO_4$, H_2S (b) $BaSO_3$, H_2S

(c) $BaSO_4$, SO_2 (d) $BaSO_3$, SO_2

60. On heating potassium permanganate, one of the following compounds is not obtained CET 2015

(a) $2MnO_2$ (b) O_2

(c) K_2MnO_4 (d) MnO

61. The man and permanganate ions are tetrahedral due to NEET 2019

(a) The Л bonding involves overlap of d orbitals of oxygen with d orbitals of manganese

(b) The Л bonding involves overlap of p orbitals of oxygen with d orbitals of manganese

(c) There is no Л bonding

(d) The Л bonding involves overlap of p orbitals of oxygen with p orbitals of manganese

62. Name the gas that can readily decolorize acidified $KMnO_4$ solution NEET 2017

(a) SO_2 (b) NO_2

(c) P_2O_5 (d) CO_2

63. The incorrect statement is

(a) In manganate and permanganate ions the Л bonding takes place by overlap of p orbitals of oxygen and d orbitals of manganese

(b) Manganate ion is green in color and permanganate ion is purple in color

(c) Manganate and permanganate ions are paramagnetic

(d) Manganate and permanganate ions are tetrahedral

64. The color and magnetic nature of manganate ion MnO_4^{-2} is

(a) Green, paramagnetic (b) Purple, diamagnetic

(c) Green, diamagnetic (d) Purple, paramagnetic

65. Which of the following ions contains a bridging oxygen atom?

(a) MnO_4^- (b) MnO_4^{2-}

(c) CrO_4^{2-} (d) $Cr_2O_7^{2-}$

THE INNER TRANSITION ELEMENTS (f-BLOCK)

66. Reduction in atomic size with increase in atomic number is a characteristic of elements of

(a) high atomic masses (b) *d*-block

(c) *f*-block (d) radioactive series

Lanthanoids

67. The radius of La^{3+}(Z=57) is 106pm. Which one of the following given values will be closest to the radius of Lu^{3+}(Z=71)?

(a) 160pm (b) 140 pm

(c) 106 pm (d) 85pm

68. Non – lanthanoid atom is

(a) La (b) Lu

(c) Pr (d) Pm

Electronic Configuration

69. The electronic configurations of Eu(Atomic No.63), Gd(Atomic No.64) and Tb(Atomic No.65) are

(a) $[Xe]4f^7 6s^2$ $[Xe]4f^8 6s^2$ and $[Xe]4f^8 5d^1 6s^2$ (b) $[Xe]4f^8 5d^1 6s^2$ $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^8 6s^2$

(c) $[Xe]4f^6 5d^1 6s^2$ $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^8 5d^1 6s^2$ (d) $[Xe]4f^7 6s^2$ $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^9 6s^2$

70. Gadolinium belongs to 4*f* series. Its atomic number is 64. Which of the following is the correct electronic configuration of gadolinium?

(a) $[Xe]4f^8 6d^2$ (b) $[Xe]4f^9 5s^1$

(c) $[Xe]4f^7 5d^1 6s^2$ (d) $[Xe]4f^6 5d^2 6s^2$

Lanthanoid contraction

71. Lanthanoid contraction can be observed in

(a) La (b) Lu

(c) Gd (d) Ac

72. Which of the following factors may be regarded as the main cause of lanthanoid contraction?

(a) Greater shielding of 5*d* electrons by 4*f* – electrons

(b) Poorer shielding of 5*d* electrons by 4*f* – electrons

(c) Effective shielding of one of 4*f* – electrons by another in the subshell

(d) Poorer shielding of one of 4*f* – electrons by another in the subshell

73. Reason of lanthanoid contraction is:-

(a) Negligible screening effect of ‘f ’ orbitals. (b) Increasing nuclear charge

(c) Decreasing nuclear change. (d) Decreasing screening effect

74. Which one of the following statements concerning Lanthanoids elements is false?

(a) Lanthanoids are separated from one another by ion exchange methods.

(b) The ionic radii of trivalent Lanthanoids steadily increase with increase in atomic number.

(c) All Lanthanoids are highly dense metals.

(d) Most typical oxidation of Lanthanoids is +3

Consequences of Lanthanoid contraction

75. Identify the incorrect statement among the following:

(a) Zirconium and Hafnium have identical radii of 160pm and 159 pm, respectively as a consequence of lanthanoid contraction.

(b) Lanthanoids reveal only +3 oxidation state.

(c) The lanthanoid ions other than f^0 type and the f^{14} type are are all paramagnetic.

(d) lanthonoid contraction.

76. Because of Lanthanoid contraction, which of the following pairs of elements have nearly same atomic radii? (Numbers in the parenthesis are atomic numbers).

(a) Zr(40) and Nb(41) (b) Zr(40) and Hf(72)

(c) Zr(40) and Ta(73) (d) Ti(22) and Zr(40)

Oxidation States

77. Which of the following oxidation states is the most common among the lanthanoids?

(a) +3 (b) +4

(c) +2 (d) +5

78. Although +3 is the characteristic oxidation state for lanthanoids but cerium also shows +4 oxidation state +4 oxidation state because----------

(a) it has variable ionization enthalpy

(b) it has a tendency to attain noble gas configuration

(c) it has a tendency to attain f^0 configuration

(d)it resembles Pb^{4+}

79. Which of the following exhibit only +3 oxidation state?

(a) U (b) Th

(c) Ac (d) Pa

80. Cerium (Z=58) is an important member of lanthanoids. Which of the following statements about cerium is not correct

(a) The common oxidation states of Ce are +3 and +4

(b) +3 oxidation state is more stable than +4 oxidation state.

(c) +4 oxidation state is not known in solution.

(d) Ce(IV) acts as an oxidizing agent.

81. The +3 ion of which one of the following has half-filled 4fsubshell?

(a) La (b) Lu

(c) Gd (d) Ac

82. The reaction for the stability of Gd^{3+} ion is

(a) 4*f* subshell —half filled

(b) 4*f* subshell —completely filled

`(c) Possesses the general electronic configuration of noble gases

(d) 4*f* subshell —completely filled

83. The correct order of ionic radii of Y^{3+}, La^{3+}, Eu^{3+}, and Lu^{3+} is (Atomic nos Y=39, La=57, Eu=63, Lu=71)

(a) $La^{3+}<Eu^{3+}< Lu^{3+} <Y^{3+}$ (b) $Y^{3+} <La^{3+} <Eu^{3+}< Lu^{3+}$

(c) $Y^{3+}< Lu^{3+}< Eu^{3+}< La^{3+}$ (d) $Lu^{3+}<Eu^{3+}< La^{3+} <Y^{3+}$

General Characteristics

84. Which one of the following elements shows maximum number of different oxidation states in its compounds

(a) Eu (b) La

(c) Gd (d) Am

85. Which of the following is the use of mischmetall?

(a) In bullets (b) In lighter flint

(c) As catalyst in petroleum cracking (d) Both 1) and 2)

86. What is the approximate percentage of lanthanoid metal in mischmetall?

(a) 90% (b) 20%

(c) 5% (d) 95%

87. Gun metal is an alloy of

(a) Cu and AI (b) Cu and Sn

(c) Cu, Zn and Sn (d) Cu, Zn and Ni

88. Which of the following lanthanoid ions is diamagnetic? (At nos. Ce=58, Sm=62, Eu=63 Yb=70)

(a) Sm^{2+} (b) Eu^{2+}

(c) Yb^{2+} (d) Ce^{2+}

89. Among the lanthanoids, the one obtained by synthetic method is

(a) Lu (b) Pm

(c) Pr (d) Gd

90. The approximate percentage of iron in mischmetal is

(a) 10 (b) 20

(c) 50 (d) 5

91. The heaviest atom amongst the following is

(a) Uranium (b) Radium

(c) Europium (d) Cerium

The Actinoids

92. The reason for greater range of oxidation states in actinoids is attributed to:

(a) Actinoid contraction. (b) 5*f*, 6*d* and 7*s* levels having comparable energies.

(c) 4*f*, and 5*d* levels being clse in energies. (d)The radioactive nature of actinoids

93. Identify the incorrect statement among the following:

(a) Most of the trivalent Lanthanoid ions are colorless in the solid state

(b) Lanthanoid are good conductors of heat and electricity

(c) Actinoids are highly reactive metals, especially when finely divided

(d) Actinoid contraction is greater for element to element than lanthanoid contraction

94. Identify the incorrect statement among the following:

(a) 4*f* and 5*f* orbitals are equally shielded.

(b) *d*- Block elements show irregular and erratic chemical properties among themselves

(c) La and Lu have partially filled d-orbitals and no other partially filled orbitals.

(d) The chemistry of various lanthanoids is very similar.

Some Applications of d- and f-Block Elements

95. Which one of the following is coinage metal?

(a) Zn (b) Cu

(c) Sn (d)Pb

96. Which of the following reactions represents “developing” in photography?

(a) $AgNO_3+NaBr—>AgBr + NaNO_3$ (b) $AgBr+2Na_2S_2O_3—>Na_3[Ag(S_3O_3)_2]+NaBr$

(c) $AgBr+hv—>AgBr^*$ (d)$C_6H_4(OH)_2++2AgBr^*—>C_6H_4O_2+2HBr+2Ag$

97. Brass is an alloy of

(a) Zn and Sn (b) Zn and Cu

(c) Cu, Zn and Sn (d) Cu and Sn

ANSWER KEY:

1. d	2. a	3.b	4.c	5.c	6.b	7.b	8.a	9.d	10.c
11.d	12.a	13.c	14.b	15.a	16.a	17.c	18.d	19.b	20.b
21.a	22.a	23.b	24.d	25.d	26.c	27.b	28.a	29.a	30.a
31.b	32.c	33.d	34.c	35.c	36.a	37.b	38.a	39.a	40.c
41.c	42.c	43.a	44.b	45.b	46.c	47.a	48.d	49.b	50.c
51.a	52.b	53.c	54.d	55.a	56.d	57.a	58.b	59.d	60.d
61.b	62.a	63.c	64.a	65.d	66.c	67.d	68.a	69.d	70.c
71.b	72.d	73.a	74.b	75.b	76.b	77.a	78.b	79.c	80.c
81.c	82.a	83.c	84.d	85.d	86.d	87.c	88.c	89.b	90.d
91.a	92.b	93.a	94.a	95.b	96.d	97.b			

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